

## Exam: Chi-Square Tests

*Multiple Choice (Single Select) - 4 Answer Options*

### 1. What does the chi-square test assess?

- A) Correlations
- B) Variance differences
- C) Mean differences
- D) Relationships between categorical variables

### 2. Which assumption applies to the chi-square test?

- A) Expected frequencies > 5 per cell
- B) Metric data
- C) Normality
- D) Equal group sizes

### 3. What is Cramer's V?

- A) A normality test
- B) A measure of effect size for chi-square
- C) A correlation measure
- D) A post-hoc test

### 4. When do you use Fisher's exact test?

- A) With small samples or expected frequencies < 5
- B) With metric data
- C) With large samples
- D) With more than 4 cells

### 5. What is the phi coefficient?

- A) Effect size for t-tests
- B) The chi-square value
- C) Special case of Cramer's V for 2x2 tables
- D) The p-value

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**6. What does a significant chi-square test mean?**

- A) The variables are causally related
- B) The variables are related
- C) The variances are homogeneous
- D) The means differ

**7. How is chi-square calculated?**

- A) Sum (Observed - Expected)
- B) Product of row and column sums
- C)  $\text{Sum (Observed - Expected)}^2 / \text{Expected}$
- D) Mean of differences

**8. How many degrees of freedom does a 3x4 table have?**

- A) 2
- B) 12
- C) 7
- D) 6

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## Solutions

1: D

2: A

3: B

4: A

5: C

6: B

7: C

8: D